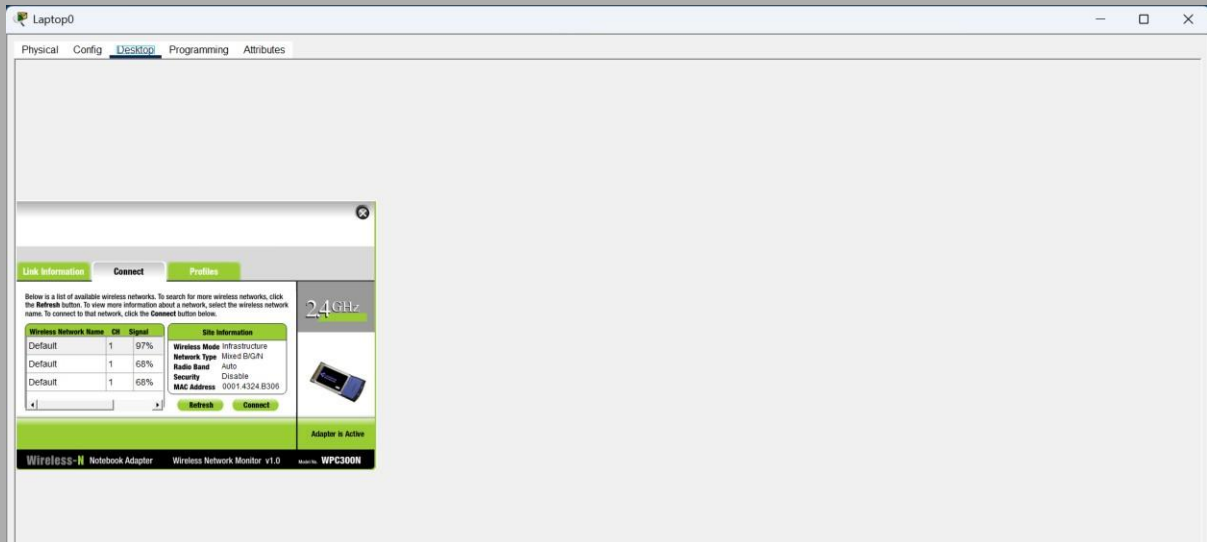


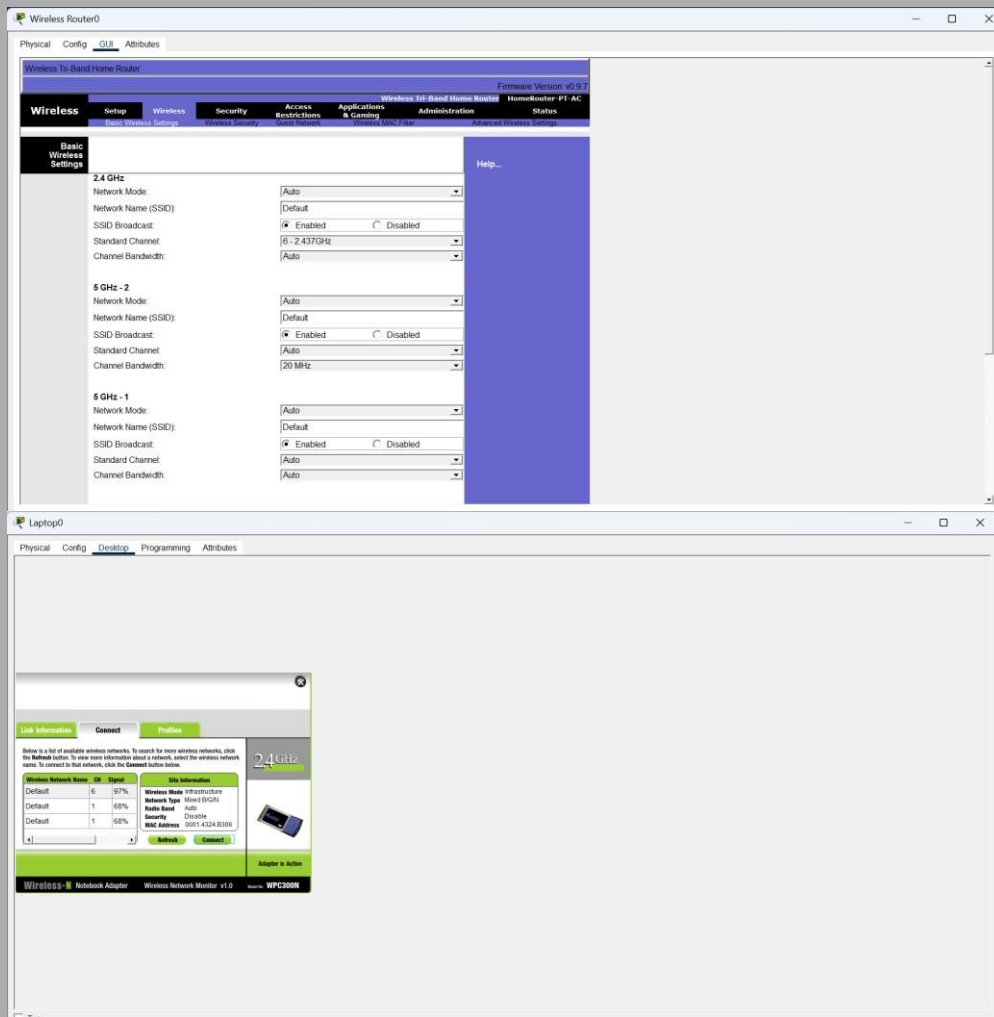
Wireless standard and configuration

1. Setup a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

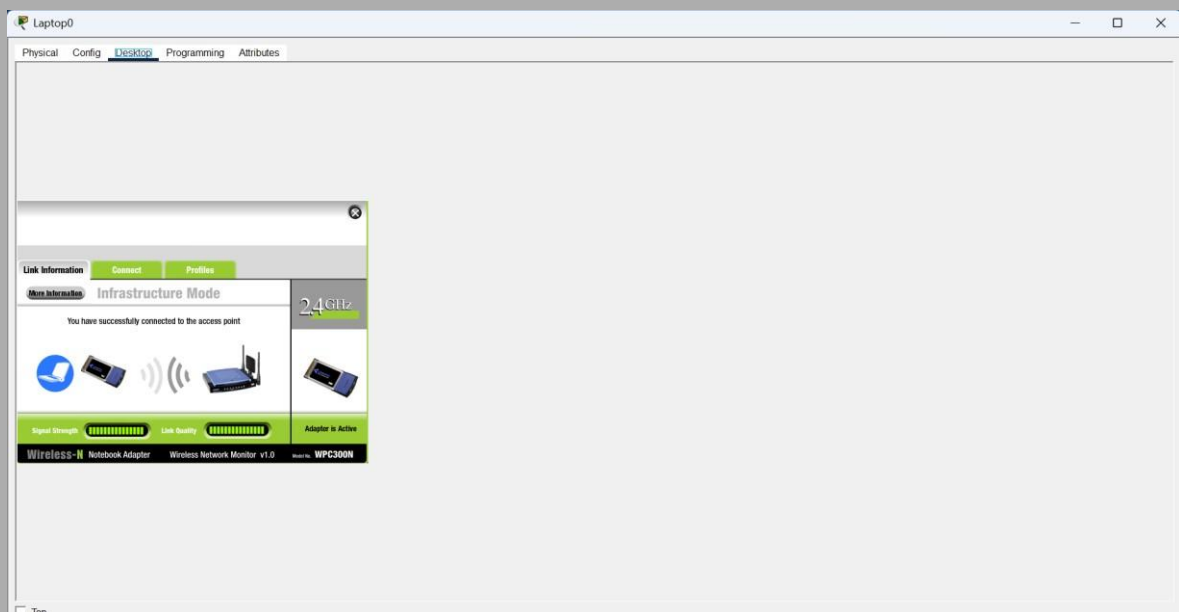


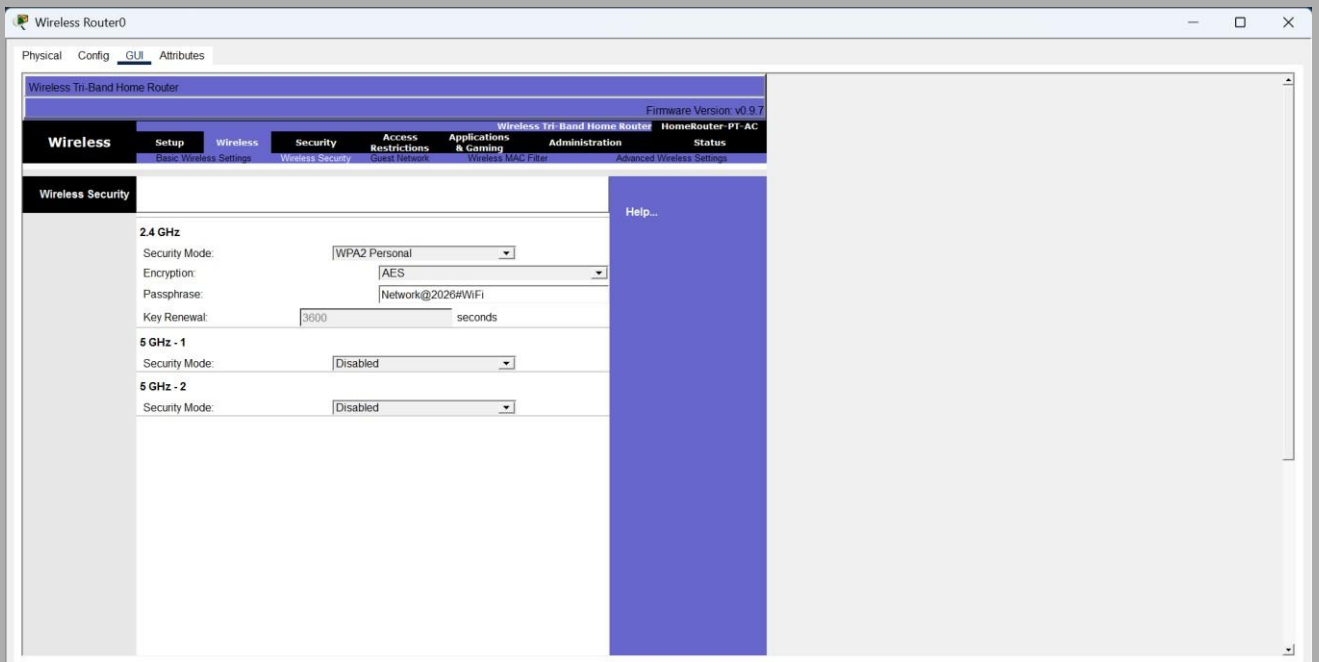
2. Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks.

Hint: Look for overlapping channels and pick one with the least overlap.



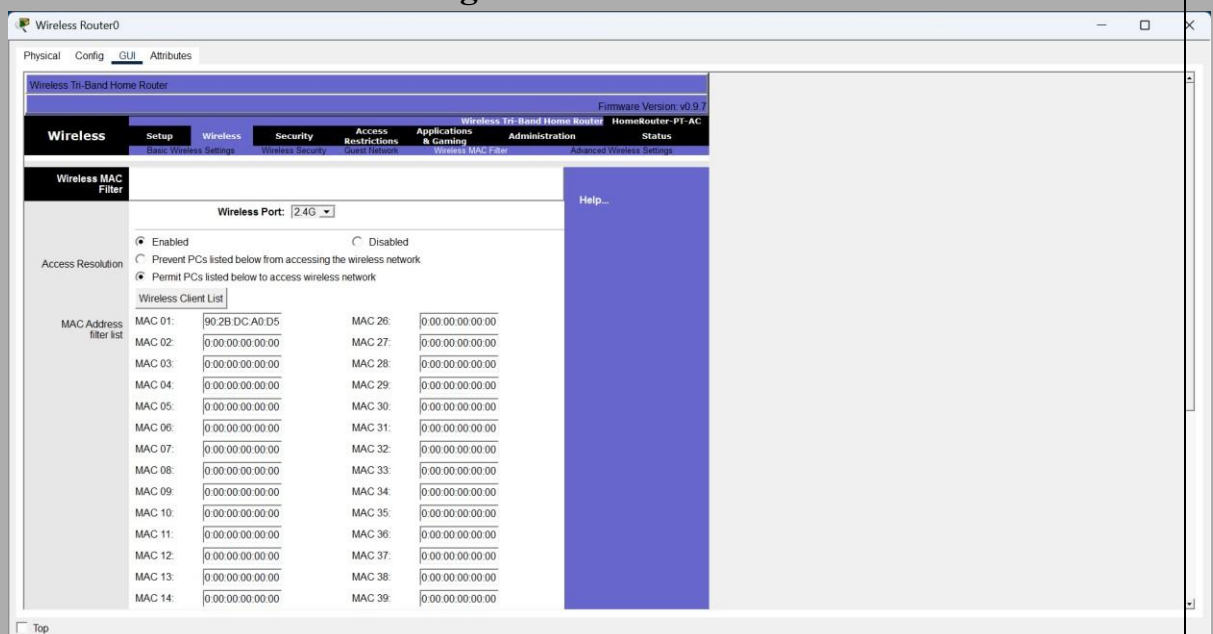
3. Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active

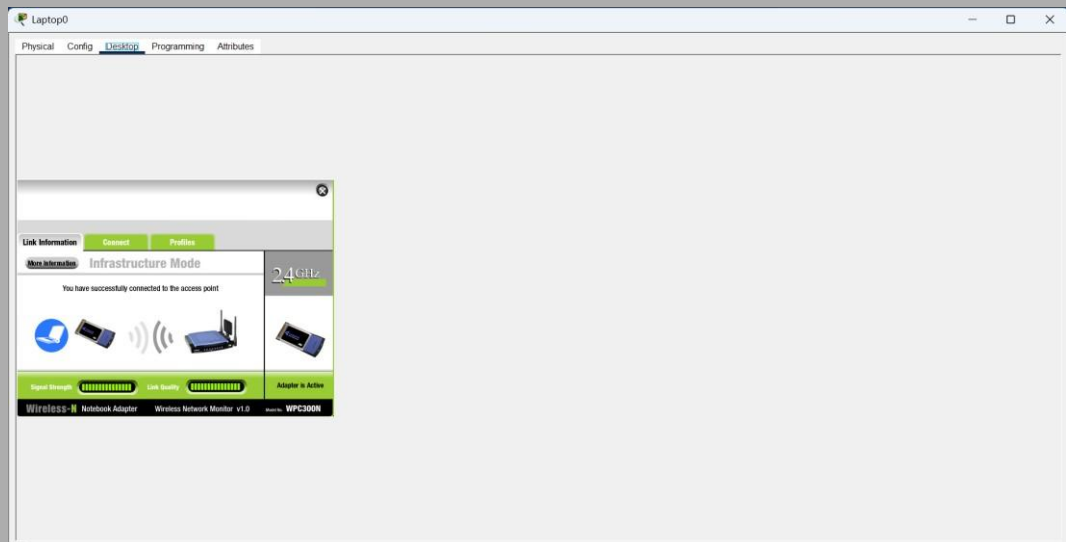




4. Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result.

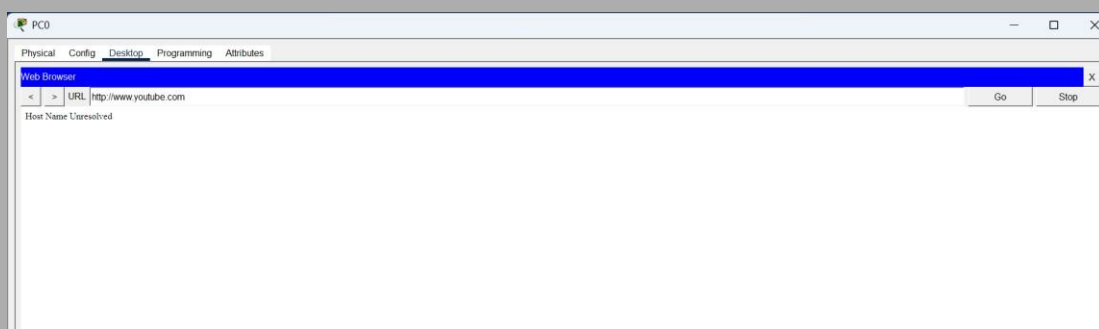
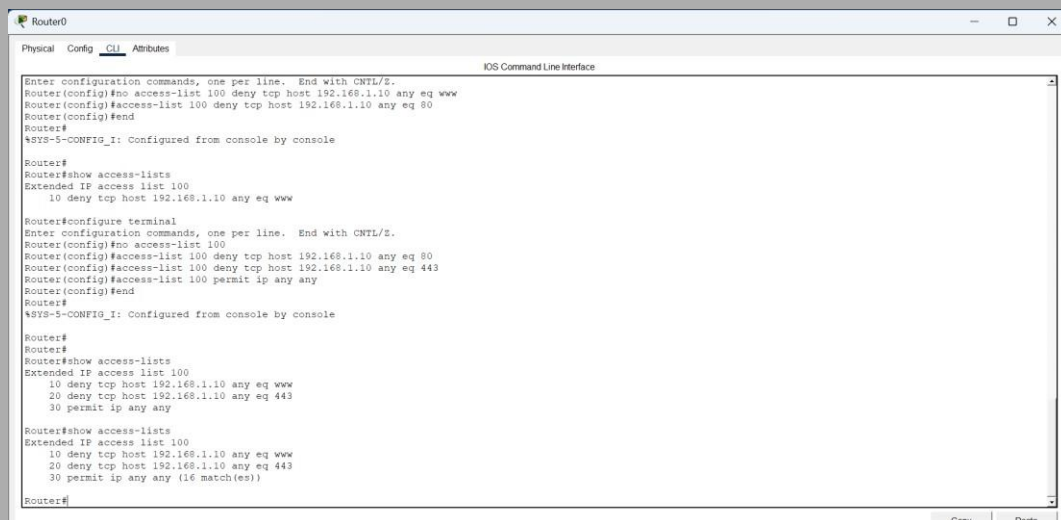
Hint: You can find your device's MAC address in the network settings. ○





The laptop's MAC address was added to the router's allowed MAC list. The laptop was successfully connected to the wireless network. A device whose MAC address was not added to the allowed list could not access the wireless network.

5. Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.



I created an extended ACL on the router to block HTTP (port 80) and HTTPS (port 443) traffic from the PC. The ACL was applied to the router interface, and the configuration was verified using the show access-lists command. When I tried to access YouTube from the PC, the webpage did not load, showing that the access restriction was configured.